Amazon Sales Data Analysis Using SQL Project Presentation

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Overview Of Amazon Sales Data

- The data consists of sales record of three cities/branch in Myanmar which are Naypyitaw, Yangon, Mandalay which took place in first quarter of year 2019. The data consists of 1000 rows and 17 columns. The objective of this project is to understand sales performance, customer behavior, product trends, and revenue contribute on using SQL.

Objective Of Amazon Sales Data

- The main objectives of this project are:
- To analyze overall sales performance of Amazon
- To identify top-performing and low-performing product lines
- To study customer behavior based on gender and customer type
- To analyze sales trends over time (monthly, daily, hourly)
- To calculate total revenue and profit
- To derive key business insights for decision-making

Amazon Sales Data Description

```
4 •   select * from amazon;
5
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Invoice ID	Branch	City	Customer type	Gender	Product line	Unit price	Quantity	Tax 5%	Total	Date	Time	Payment	cogs	gross perce
750-67-8428	A	Yangon	Member	Female	Health and beauty	74.69	7	26.1415	548.9715	2019-01-05	13:08:00	Ewallet	522.83	4.761
226-31-3081	C	Naypyitaw	Normal	Female	Electronic accessories	15.28	5	3.82	80.22	2019-03-08	10:29:00	Cash	76.4	4.761
631-41-3108	A	Yangon	Normal	Male	Home and lifestyle	46.33	7	16.2155	340.5255	2019-03-03	13:23:00	Credit card	324.31	4.761
123-19-1176	A	Yangon	Member	Male	Health and beauty	58.22	8	23.288	489.048	2019-01-27	20:33:00	Ewallet	465.76	4.761
373-73-7910	A	Yangon	Normal	Male	Sports and travel	86.31	7	30.2085	634.3785	2019-02-08	10:37:00	Ewallet	604.17	4.761
699-14-3026	C	Naypyitaw	Normal	Male	Electronic accessories	85.39	7	29.8865	627.6165	2019-03-25	18:30:00	Ewallet	597.73	4.761
355-53-5943	A	Yangon	Member	Female	Electronic accessories	68.84	6	20.652	433.692	2019-02-25	14:36:00	Ewallet	413.04	4.761
315-22-5665	C	Naypyitaw	Normal	Female	Home and lifestyle	73.56	10	36.78	772.38	2019-02-24	11:38:00	Ewallet	735.6	4.761
665-32-9167	A	Yangon	Member	Female	Health and beauty	36.26	2	3.626	76.146	2019-01-10	17:15:00	Credit card	72.52	4.761

amazon 1 × Read Only

Tools and Technologies Used

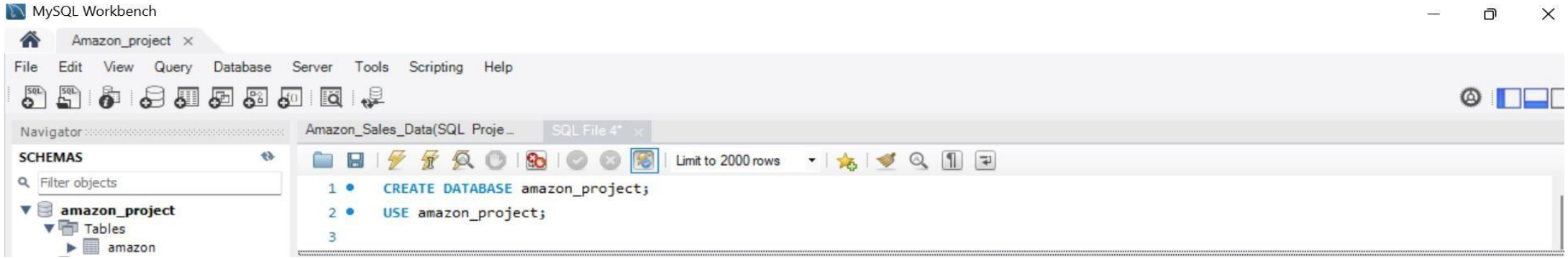
- MySQL Database
- MySQL Workbench

- SQL (Structured Query Language)
- CSV Dataset

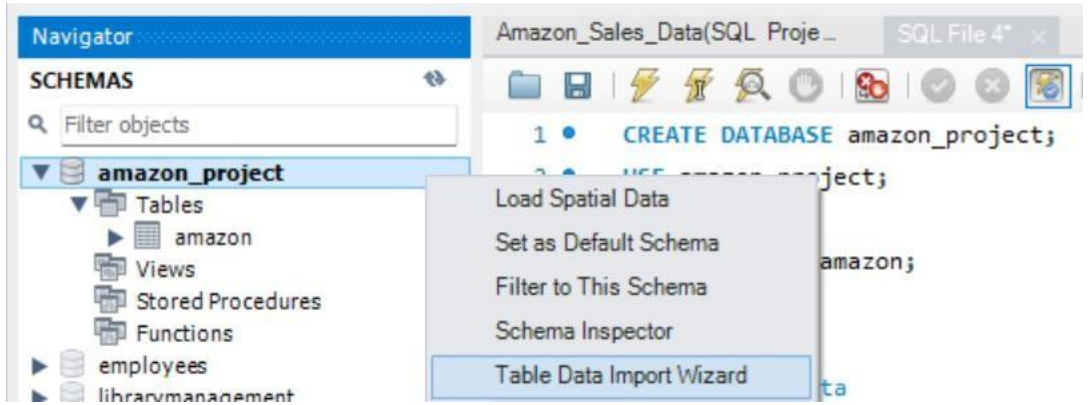
Database and Table Creation

- A database named amazon_project was created in MySQL Workbench. The CSV file was imported into a table named amazon using the Table Data Import Wizard.

    1. Created a database named Amazon Project in MySQL.



    2. Import the data table named Amazon using table data import wizard.



3. Checking the NULL values and datatypes of the columns of demo amazon table.

```
# Checking the datatypes of the columns of demo amazon table.
describe amazon;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
Invoice ID	text	YES		NULL	
Branch	text	YES		NULL	
City	text	YES		NULL	
Customer type	text	YES		NULL	
Gender	text	YES		NULL	
Product line	text	YES		NULL	
Unit price	double	YES		NULL	
Quantity	int	YES		NULL	

Result 1

```
# Checking the NULL values amazon table.
select count(*) as count_of_null_values from amazon where null;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

count_of_null_values
0

4. Display the first 5 records from the Amazon table.

```
15 •  SELECT * FROM amazon LIMIT 5;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

Invoice ID	Branch	City	Customer type	Gender	Product line	Unit price	Quantity	Tax 5%	Total	Date	Time	Payment	cogs	gross mar percenta
750-67-8428	A	Yangon	Member	Female	Health and beauty	74.69	7	26.1415	548.9715	2019-01-05	13:08:00	Ewallet	522.83	4.761904
226-31-3081	C	Naypyitaw	Normal	Female	Electronic accessories	15.28	5	3.82	80.22	2019-03-08	10:29:00	Cash	76.4	4.761904
631-41-3108	A	Yangon	Normal	Male	Home and lifestyle	46.33	7	16.2155	340.5255	2019-03-03	13:23:00	Credit card	324.31	4.761904
123-19-1176	A	Yangon	Member	Male	Health and beauty	58.22	8	23.288	489.048	2019-01-27	20:33:00	Ewallet	465.76	4.761904
373-73-7910	A	Yangon	Normal	Male	Sports and travel	86.31	7	30.2085	634.3785	2019-02-08	10:37:00	Ewallet	604.17	4.761904

Answering Business Questions:

1. What is the count of distinct cities in the dataset.

```
16    # 1. What is the count of distinct cities in the dataset.
17 •  select count(distinct(city)) from amazon;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

count(distinct(city))
3

2. For each branch, what is corresponding city.

```
# 2. For each branch, what is corresponding city.
select distinct city,branch from amazon;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

city	branch
Yangon	A
Naypyitaw	C
Mandalay	B

3. How many records are present in the dataset. What are the different cities where Amazon operates.

```
# How many records are present in the dataset.
SELECT COUNT(*) AS total_records FROM amazon;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

total_records
1000

4. What are the different cities where Amazon operates.

```
# What are the different cities where Amazon operates.
SELECT DISTINCT city FROM amazon;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

city
Yangon
Naypyitaw
Mandalay

5. Which city generates the highest revenue.

```
# Which city generates the highest revenue?
select City, SUM(Total) AS City_Sales from amazon group by City;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

City	City_Sales
Yangon	106200.3705000001
Naypyitaw	110568.70649999994
Mandalay	106197.67199999996

6. Which branch generates the highest revenue.

```sql
# Which branch generates the highest revenue?
select Branch, SUM(Total) AS Branch_Revenue from amazon
group by Branch order by Branch_Revenue DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Branch	Branch_Revenue
C	110568.70649999994
A	106200.3705000001
B	106197.67199999996

7. What are the different product lines available.

```sql
# What are the different product lines available.
SELECT DISTINCT `Product line` FROM amazon;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Product line
Health and beauty
Electronic accessories
Home and lifestyle
Sports and travel
Food and beverages
Fashion accessories

8. Which product line generates the highest revenue.

```sql
# Which product line generates the highest revenue.
SELECT `Product line`, SUM(total) AS revenue FROM amazon GROUP BY `Product line` ORDER BY revenue DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Product line	revenue
Food and beverages	56144.844000000005
Sports and travel	55122.826499999996
Electronic accessories	54337.531500000005
Fashion accessories	54305.895
Home and lifestyle	53861.91300000001
Health and beauty	49193.739000000016

9. What is the sales distribution by customer type.

```sql
# What is the sales distribution by customer type.
SELECT `Customer type`, SUM(total) AS sales FROM amazon GROUP BY `Customer type`;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Customer type	sales
Member	164223.44400000002
Normal	158743.30500000005

10. What is the average spending by gender.

```sql
# What is the average spending by gender.
SELECT gender, ROUND(AVG(total),2) AS avg_spending FROM amazon GROUP BY gender;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

gender	avg_spending
Female	335.1
Male	310.79

Key Findings and Insights

- Food and Beverages is the highest revenue-generating product line
- Electronic Accessories has the highest sales volume
- January is the highest revenue month
- Naypyitaw (Branch C) contributes the highest revenue
- Members generate more revenue than normal customers
- Female customers contribute higher revenue compared to male customers
- Afternoon and Saturdays are peak sales periods

Conclusion

- This project successfully analyzed Amazon sales data using SQL. By performing product, sales, customer, time-based, and profit analysis, meaningful insights were derived. These insights can help businesses optimize product strategies, improve customer targeting, and enhance overall sales performance.

Future Scope

- Integration with visualization tools such as Power BI or Tableau
- Advanced analysis using CTEs and subqueries
- Predictive analysis using historical sales data
- Automation of reporting dashboards

Amazon Sales Data Analysis Project

Overview

This project analyzes Amazon sales data of three cities in Myanmar: Yangon, Mandalay, and Naypyitaw for the first quarter of 2019.

Objective

The main objective is to understand sales performance and customer behavior using data analysis and visualization.

In [5]:
```
import pandas as pd

df = pd.read_csv("Amazon.csv")
df.head()
```

Out[5]:

	Invoice ID	Branch	City	Customer type	Gender	Product line	Unit price	Quantity	Tax 5%	
0	750-67-8428	A	Yangon	Member	Female	Health and beauty	74.69	7	26.1415	54
1	226-31-3081	C	Naypyitaw	Normal	Female	Electronic accessories	15.28	5	3.8200	8
2	631-41-3108	A	Yangon	Normal	Male	Home and lifestyle	46.33	7	16.2155	34
3	123-19-1176	A	Yangon	Member	Male	Health and beauty	58.22	8	23.2880	48
4	373-73-7910	A	Yangon	Normal	Male	Sports and travel	86.31	7	30.2085	63

Group data by City

In [6]:
```
# Group data by 'City' and sum the 'Total' sales
city_sales = df.groupby("City")["Total"].sum()
```

```
# Display city-wise total sales
city_sales
```

Out[6]:
```
City
Mandalay     106197.6720
Naypyitaw    110568.7065
Yangon       106200.3705
Name: Total, dtype: float64
```

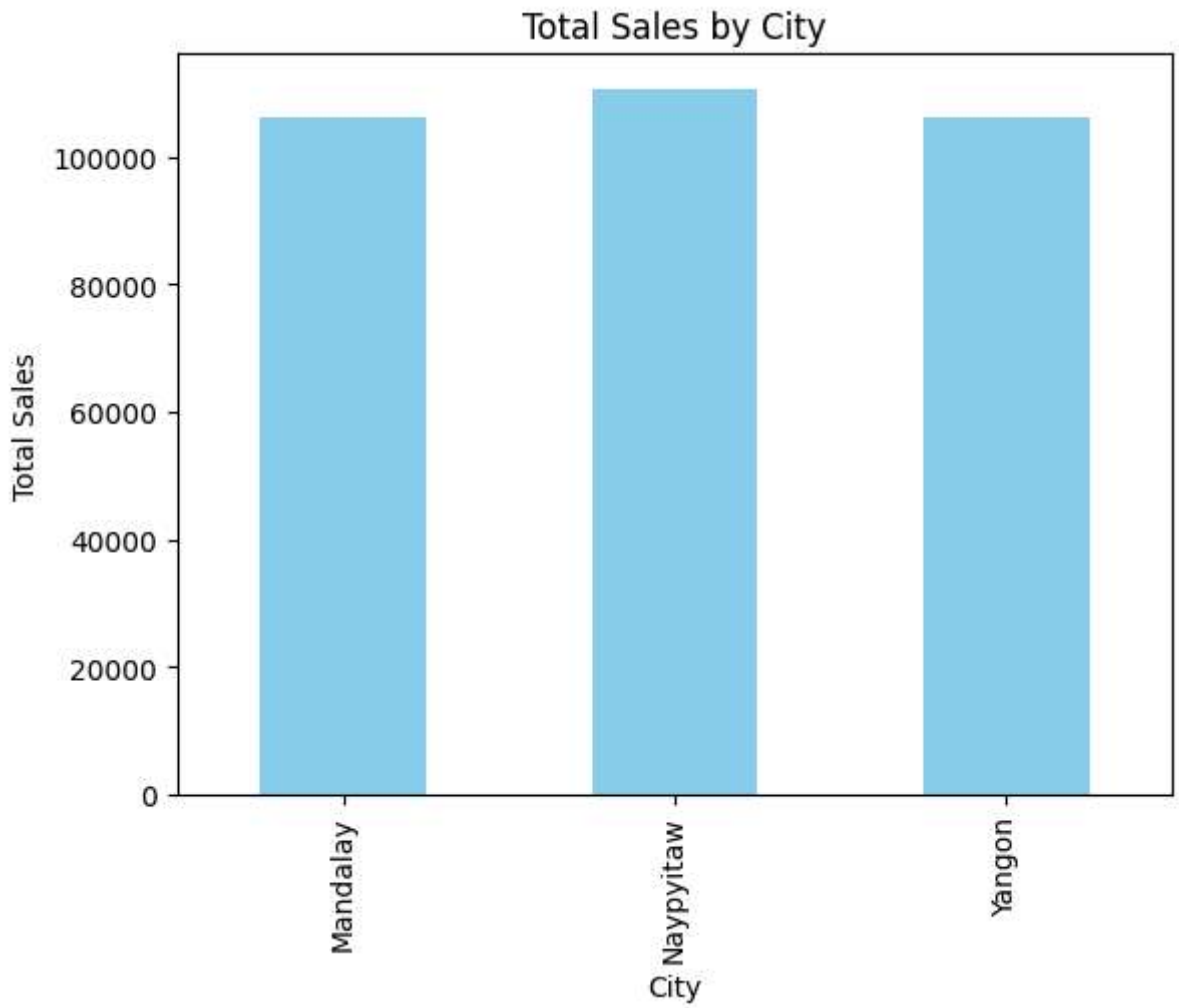
Bar Chart (City-wise Sales)

In [7]:
```
# Import matplotlib for plotting
import matplotlib.pyplot as plt

# Plot bar chart for city-wise total sales
city_sales.plot(kind="bar", color="skyblue")

# Add title and labels
plt.title("Total Sales by City")
plt.xlabel("City")
plt.ylabel("Total Sales")

# Show the plot
plt.show()
```



Payment Method Pie Chart

Objective:

Show which payment method customers use the most.

In [ ]:
```
## Count number of transactions per payment method
```

In [8]:
```
# Count number of transactions for each payment method
payment_count = df["Payment"].value_counts()

# Display the count
payment_count
```

Out[8]:
```
Payment
Ewallet        345
Cash           344
Credit card    311
Name: count, dtype: int64
```

Pie Chart (Payment Method)

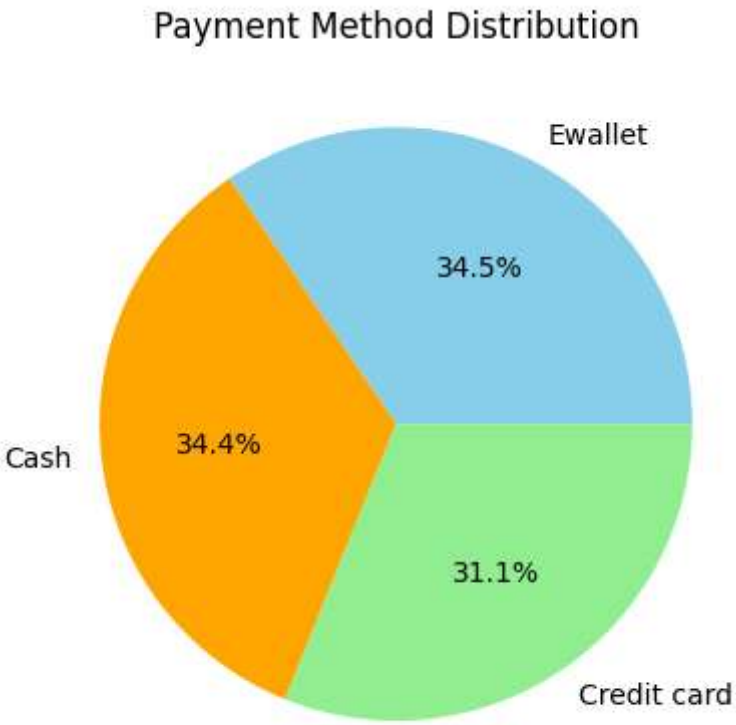
In [9]:
```
# Import matplotlib (agar pehle import nahi kiya)
import matplotlib.pyplot as plt

# Pie chart for payment method distribution
payment_count.plot(kind="pie", autopct="%1.1f%%", colors=["skyblue", "orange", "lig

# Add title
plt.title("Payment Method Distribution")

# Remove y-axis label
plt.ylabel("")

# Show the plot
plt.show()
```



Product Line Revenue Bar Chart

Objective:

Show which product line generates the highest total revenue.

Group data by Product Line

In [10]:
```
# Group data by 'Product line' and sum total revenue
product_sales = df.groupby("Product line")["Total"].sum()

# Display product line revenue
product_sales
```

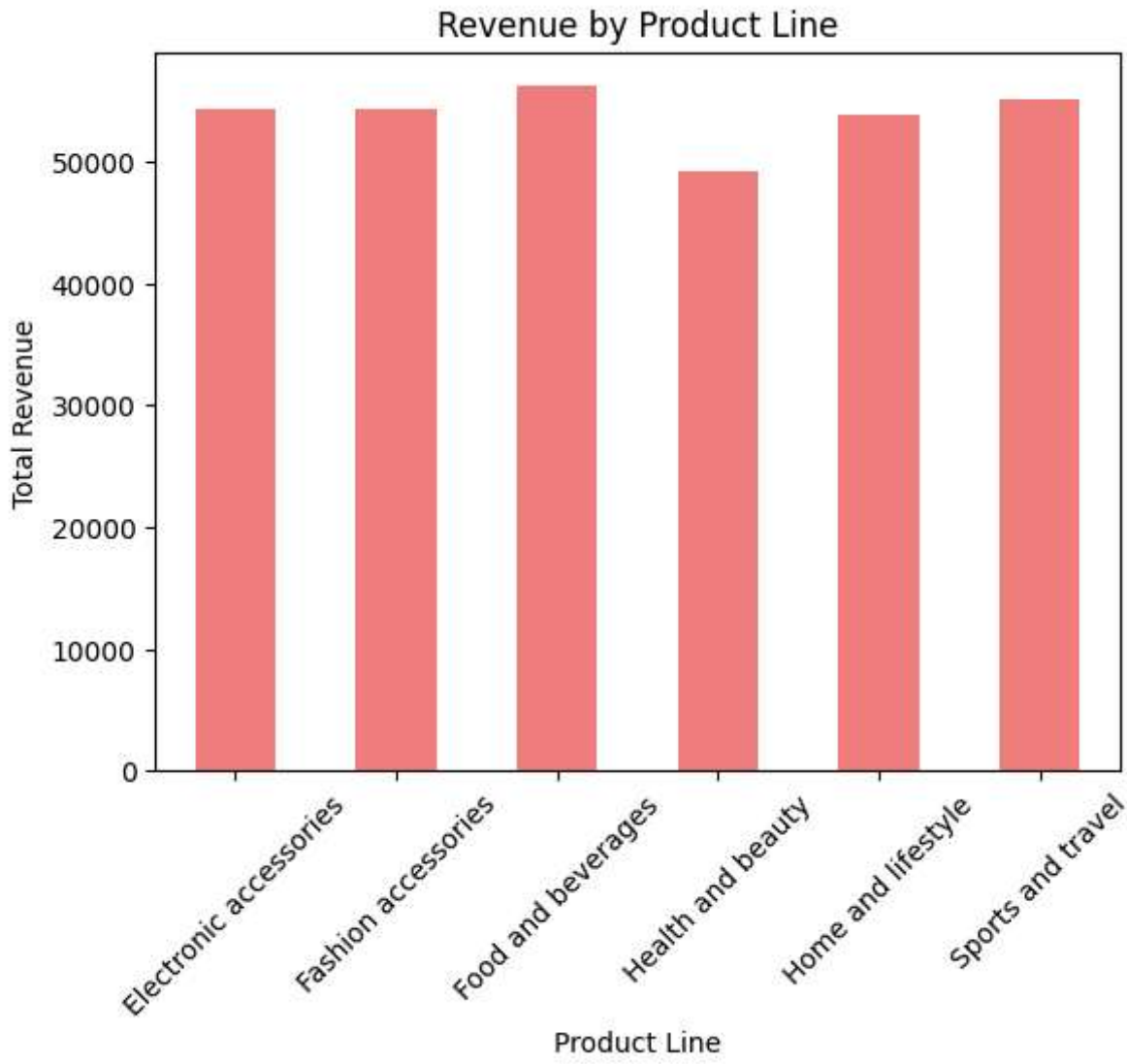
Out[10]:
```
Product line
Electronic accessories    54337.5315
Fashion accessories       54305.8950
Food and beverages        56144.8440
Health and beauty         49193.7390
Home and lifestyle        53861.9130
Sports and travel         55122.8265
Name: Total, dtype: float64
```

Bar Chart for Product Line Revenue

In [12]:
```
# Plot bar chart for product line revenue
product_sales.plot(kind="bar", color="lightcoral")

# Add title and labels
plt.title("Revenue by Product Line")
plt.xlabel("Product Line")
plt.ylabel("Total Revenue")
plt.xticks(rotation=45)  # Rotate x-axis labels for better visibility

# Show the plot
plt.show()
```



Monthly Sales Trend (Line Chart)

Extract Month from Date

In [14]:
```
# Convert 'Date' column to datetime format (day first)
df["Date"] = pd.to_datetime(df["Date"], dayfirst=True)

# Extract month number
df["Month"] = df["Date"].dt.month

# Check first 5 rows
df[["Date","Month"]].head()
```

Out[14]:

	Date	Month
0	2019-01-05	1
1	2019-03-08	3
2	2019-03-03	3
3	2019-01-27	1
4	2019-02-08	2

Line Chart for Monthly Sales

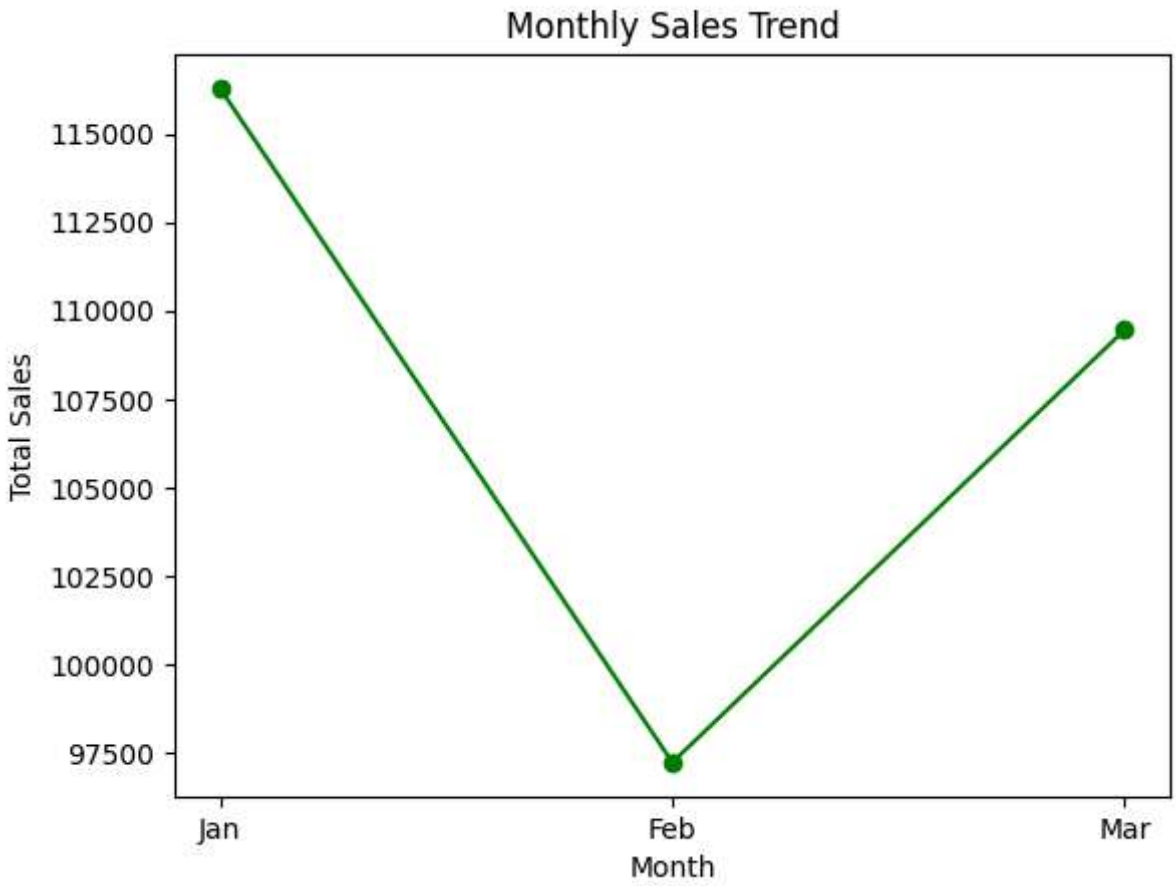
In [15]:

```
# Group total sales by month
monthly_sales = df.groupby("Month")["Total"].sum()

# Plot line chart
monthly_sales.plot(kind="line", marker="o", color="green")

# Add title and labels 3
plt.title("Monthly Sales Trend")
plt.xlabel("Month")
plt.ylabel("Total Sales")
plt.xticks([1,2,3], ["Jan","Feb","Mar"])  # Replace month numbers with names

# Show the plot
plt.show()
```



Insight: Total Sales by City

- Naypyitaw has the highest total sales.
- Yangon and Mandalay have similar sales.

Insight: Total Sales by Payment Method

- Ewallet is the most preferred payment method.
- Cash and Credit Card are used less compared to Ewallet.

Insight: Total Sales by Product Line

- Health and Beauty is the top selling product line.
- Food and Beverages and Sports and Travel also perform well.

Insight: Total Sales by Customer Type

- Member customers generate more revenue than Normal customers.

Conclusion

- Naypyitaw is the highest revenue generating city.
- Health and Beauty is the top selling product line.
- Member customers generate more revenue than Normal customers.
- Ewallet is the most preferred payment method.

This analysis helps understand customer behavior and improve business strategy.